1 Question 1 (7 Points)

Answer the following questions:

a) Describe the *Turing Test* and explain its significance in evaluating Artificial Intelligence (AI). **[3 points]**

b) List and briefly explain *four essential capabilities* that a computer must possess to pass the Turing Test. **[4 points]**

Skriv in ditt svar här

Teckenf…

Ord: 0

Totalpoäng: 7

2 Question 2 (11 Points)

For each of the following AI environments, classify them according to the properties:
(Fully/Partially Observable, Deterministic/Stochastic, Episodic/Sequential, Static/Dynamic, Discrete/Continuous, Single/Multi-Agent)

a) A chess game **[3 points]**
b) A vacuum-cleaning robot at home **[3 points]**
c) RoboCup soccer-playing game **[5 points]**

Skriv in ditt svar här

Teckenf…

Ord: 0

Totalpoäng: 11

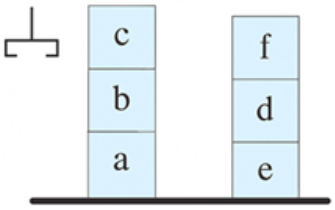
3 Question 3 (16 Points)

You are given the following description of a block world.
There are six blocks: a, b, c, d, e, and f, and a robot (agent) with a hand. Part of the initial world is described below.

- Table is clear
- Block **a** is on the table.
- Block **b** is on block **a**.
- The robot's hand is empty.

a) Based on the above information, write four different predicate logic statements that represent the current state of the world. You may choose any predicate names, but define each name in a few words. **[8 points]**

b) Using your predicates from part (a), give a sequence of actions that builds the block configuration shown in Figure 1. **[8 points]**



Skriv in ditt svar här

Teckenf...

Ord: 0

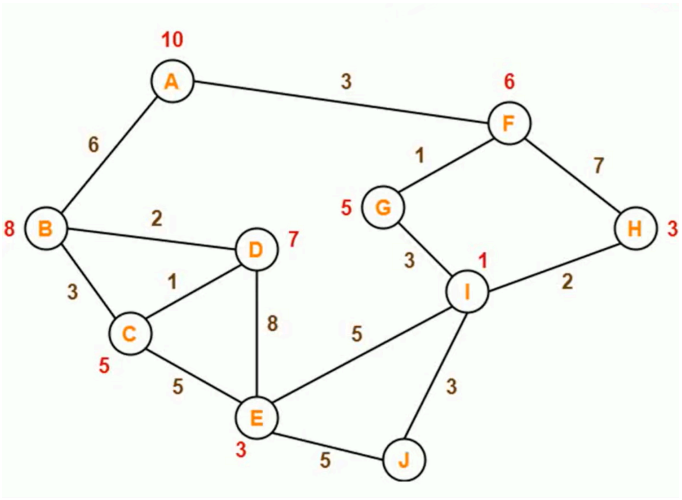
Totalpoäng: 16

4 Question 4 (13 points)

The given graph consists of 10 nodes and 14 edges. The costs are shown on the edges between the nodes and heuristic values are shown above the nodes with red color. Consider the graph (shown in the image) and design an agent using the following methods to find **the shortest path** from **A** to **J**:

a) Breadth-First Search (BFS). Show the solution in detail. **[5 Marks]**

b) The A* algorithm. Show the solution in detail. **[8 points]**



Skriv in ditt svar här

Teckenf...

Ord: 0

Totalpoäng: 13

5 Question 5 (12 points)

[12 points] Using first-order predicate logic, represent the following facts and infer the correct conclusion: **All** graduate students must take at least one AI course. John is a graduate student. CS701 is an AI course. Then, show the logical steps that lead to the inference that John must take CS701 or an equivalent AI course.

Predicate Symbols:

Predicate	Meaning
`GraduateStudent(x)`	*x* is a graduate student
`AICourse(y)`	*y* is an AI course
`Takes(x, y)`	*x* takes course *y*
`Equivalent(y1, y2)`	*y1* and *y2* are equivalent courses

Skriv in ditt svar här

Teckenf…

Ord: 0

Totalpoäng: 12

6 Question 6 (15 Points)

[Each sub-question is 3 Points] Suppose a genetic algorithm uses chromosomes of the form x = abcd with a fixed length of four genes. Each gene can be any binary number.

We want to **maximize** the function

$$f(x) = 2x^3 - 15x^2 + 36x, \quad 0 \le x \le 15, \quad x \in \mathbb{Z}.$$

1. Represent the variable x using **4-bit binary chromosomes** (e.g., $x = 0 \to 0000$, $x = 15 \to 1111$).
2. **Randomly generate 4 chromosomes** (each as a 4-bit binary string).
3. Decode each chromosome into its integer value of x.
4. Calculate the corresponding fitness value $f(x)$ for each chromosome.
5. Identify which chromosome has the highest fitness.

Skriv in ditt svar här

Teckenf...

Ord: 0

Totalpoäng: 15

7 Question 7 (10 points)

a) [5 points]

A Genetic Algorithm (GA) is used to optimize a function with the following parameters:

- Population size: $N = 40$
- Crossover probability: $p_c = 0.7$
- Mutation probability: $p_m = 0.05$
- Number of generations: $G = 80$

Assumptions:

- The GA evaluates the **fitness** of every individual **once per generation**, **after** crossover and mutation have been applied (i.e., all offspring are evaluated).
- No elitism or re-evaluation of unchanged individuals is used.
- Each fitness evaluation takes $0.02\,\text{s}$.

Tasks:

1. Compute the **total number of fitness evaluations** performed during the entire GA run.
2. Estimate the **total computation time** in seconds.

b) [5 points]

Given two parent chromosomes:

$$P_1 = (10011010), \quad P_2 = (01100101)$$

Perform a **single-point crossover** after the 4th bit.

Write down the two resulting offspring chromosomes.

Skriv in ditt svar här

Teckenf...

Ord: 0

Totalpoäng: 10

8 Question 8 (16 points)

a) (8 points) Deepfake technology and algorithmic amplification on social media are examples of AI misuse. Discuss two major *ethical risks* posed by such technologies and suggest one *regulatory or technical solution* for each.

b) (8 points) Explain the three core principles of AI Ethics — *Fairness, Accountability,* and *Transparency (FAT)* — and provide one real-world example of how each can be applied in practice.

Skriv in ditt svar här

Teckenf…

Ord: 0

Totalpoäng: 16